Lecture 6 v2:
Inference and Interpretation

25117 - Econometrics

Universitat Pompeu Fabra

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What we learned in the last lesson

- Omitted variable bias occurs when an omitted variable (a) is correlated with an included regressor and (b) is a determinant of Y.
- The multiple regression model is a linear regression model that includes multiple regressors, $X_1, \ldots, X_k$. Associated with each regressor is a regression coefficient, $\beta_1, \ldots, \beta_k$.
- The coefficient β_1 is the expected difference in Y associated with a one-unit difference in X_1, holding the other regressors constant. The other regression coefficients have an analogous interpretation.

What we learned in the last lesson

- Throughout: i.i.d. observations, exogeneity $E(u_i \mid X) = 0$, full rank, and homoskedasticity $\text{Var}(u_i \mid X) = \sigma^2$. Use the conventional covariance estimate $\widehat{V} = SER^2(X'X)^{-1}$. Exact t/F inference additionally assumes conditionally normal errors; otherwise use large-sample approximations.
- The coefficients in multiple regression can be estimated by OLS. When the four least squares assumptions are satisfied, the OLS estimators of the causal effect are unbiased, consistent, and normally distributed in large samples.
- The role of control variables is to hold constant omitted factors so that the variable of interest is no longer correlated with the error term. Properly chosen control variables can eliminate omitted variable bias in the OLS estimate of the causal effect of interest.
- Perfect multicollinearity, which occurs when one regressor is an exact linear function of the other regressors, usually arises from a mistake in choosing which regressors to include in a multiple regression. Solving perfect multicollinearity requires changing the set of regressors.

Interpreting Single Coefficients in Multiple Regression Analysis

- Tests and confidence intervals follow the same logic as in a single-regressor model.
- With conventional standard errors,

$$t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{SE(\hat{\beta}_j)} \xrightarrow{d} \mathcal{N}(0,1) \quad \text{under } H_0.$$

- With conditionally normal errors, $t_j \sim t_{n-k-1}$ exactly under H_0.
- A 95% confidence interval is

$$\hat{\beta}_j \pm t_{n-k-1;.975}\, SE(\hat{\beta}_j).$$

The critical value 1.96 is a large-sample approximation. This applies to each coefficient.

Example: Coefficients and Standard Errors

Constructed teaching example: $n = 8$, two regressors, homoskedastic OLS.

$$\widehat{Y}_i = 10 + 1X_{1i} + 1X_{2i}$$

Coefficient	Estimate	Conventional SE	t (null: zero)
Intercept	10	$\sqrt{.2} = .4472$	22.3607
X_1	1	$\sqrt{.4} = .6325$	1.5811
X_2	1	$\sqrt{.2} = .4472$	2.2361

$SSR = 8, \qquad df = 8 - 2 - 1 = 5, \qquad SER^2 = 8/5 = 1.6.$

The estimated covariance of the two slope estimates is $-.2$.

The data and a worked joint test appear below.

Tests of Joint Hypotheses

Imagine we are interested in estimating

$$TestScore_i = \beta_0 + \beta_1 frpmfracs_i + \beta_2 edfracbachplus_i + \beta_3 strs_i + u_i$$

Where $strs_i$ is the student-teacher ratio in school i. Your theory tells you that, together with the share of subsidized meals, it captures *school resources*.

You want to test whether *school resources* matter for test scores.

$$\begin{aligned} & H_0 && := \beta_1 = 0 \quad \textbf{AND} \quad \beta_3 = 0 \\ VS \quad & H_1 && := \beta_1 \neq 0 \quad \textbf{OR} \quad \beta_3 \neq 0 \quad \textbf{OR} \quad \textit{both} \end{aligned}$$

Tests of Joint Hypotheses

$$\begin{aligned} & H_0 && := \beta_1 = 0 \quad \textbf{AND} \quad \beta_3 = 0 \\ \textit{VS} \quad & H_1 && := \beta_1 \neq 0 \quad \textbf{OR} \quad \beta_3 \neq 0 \quad \textbf{OR} \quad \textit{both} \end{aligned}$$

- A joint hypothesis specifies a value for two or more coefficients, that is, it imposes a restriction on two or more coefficients.
- In general, a joint hypothesis will involve q restrictions. In the example, $q = 2$, and the two restrictions are $\beta_1 = 0$ and $\beta_3 = 0$
- A "common sense" idea is to reject if either of the individual t-statistics exceeds 1.96 in absolute value.
- But this "one at a time" test isn't valid: the resulting test rejects too often under the null hypothesis (more than 5%) !

Why can't we just test the coefficients one at a time?

- Because the rejection rate under the null isn't 5%.
- We'll calculate the probability of incorrectly rejecting the null using the "common sense" test based on the two individual t-statistics.
- For a large-sample illustration, suppose the two t-statistics are asymptotically independent standard normal under the null. Let t_1 and t_3 be

$$t_1 = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \quad \& \quad t_3 = \frac{\hat{\beta}_3}{SE(\hat{\beta}_3)}$$

- The "one at time" test would say we reject $H_0 := \beta_1 = \beta_3 = 0$ if:

$$| t_1 | > 1.96 \quad \text{AND/OR} \quad | t_3 | > 1.96$$

- What is the probability that this "one at a time" test rejects H_0 when H_0 is actually true (i.e., false positive)?

Why can't we just test the coefficients one at a time?

Large-sample illustration: assume independent standard normal limits for t_1 and t_3 under H_0.

$$\begin{aligned}
&\Pr_{H_0}(|t_1| > 1.96 \text{ or } |t_3| > 1.96) \\
&\quad = 1 - \Pr_{H_0}(|t_1| \leq 1.96 \text{ and } |t_3| \leq 1.96) \\
&\quad = 1 - \Pr_{H_0}(|t_1| \leq 1.96) \Pr_{H_0}(|t_3| \leq 1.96) \\
&\quad \approx 1 - (.95)^2 = .0975.
\end{aligned}$$

The false-positive probability is about 9.75%, above the intended 5% error probability.

The product uses independence in this illustration. In general, the rejection probability depends on the correlation between the coefficient estimates.

The F-statistic

- The F-statistic tests all q restrictions of a joint hypothesis at once.
- For $H_0 : \beta_1 = \beta_{1,0},\ \beta_2 = \beta_{2,0}$ $(q = 2)$,

$$F = \frac{1}{2}\frac{t_1^2 + t_2^2 - 2\hat{\rho}t_1 t_2}{1 - \hat{\rho}^2}, \qquad t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{SE(\hat{\beta}_j)}.$$

- Use the same conventional covariance matrix $\widehat{V} = SER^2(X'X)^{-1}$ throughout:

$$\hat{\rho} = \frac{\widehat{\mathrm{Cov}}(\hat{\beta}_1, \hat{\beta}_2)}{SE(\hat{\beta}_1)SE(\hat{\beta}_2)}.$$

- If $\hat{\rho} = 0$, $F = (t_1^2 + t_2^2)/2$; otherwise the formula accounts for dependence.
- This is exactly the restricted/unrestricted F shown below, using the same observations, hypothesis, and variance estimate. Reject for large F.

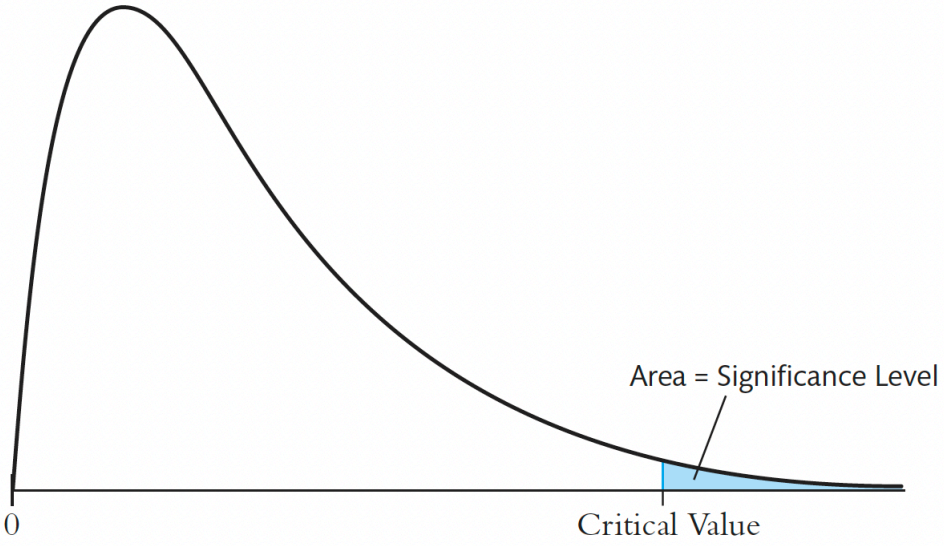
The $F_{q,n-k-1}$ distribution

q is the number of restrictions imposed by the null hypothesis H_0.

- Under the maintained assumptions and conditionally normal errors, $F \sim F_{q,n-k-1}$ exactly under H_0.
- Without normality, inference is a large-sample approximation.
- For fixed q and large denominator degrees of freedom $n-k-1$, $F_{q,n-k-1}$ approaches χ^2_q/q. Use finite degrees of freedom when needed.
- χ^2_q is the sum of q independent squared standard normal variables.
- The table gives large-sample critical values: $F_{q,\infty} = \chi^2_q/q$.

Degrees of Freedom	10%	5%	1%
1	2.71	3.84	6.63
2	2.30	3.00	4.61
3	2.08	2.60	3.78
4	1.94	2.37	3.32
5	1.85	2.21	3.02

The $F_{q,\infty}$ distribution



The F-statistic from Restricted and Unrestricted Fits

The same F-statistic can be computed by comparing two OLS fits under the maintained assumptions.

1. Fit the restricted model, imposing H_0.
2. Fit the unrestricted model, which permits both the null and the alternative.
3. Compare their fit: if relaxing the restrictions improves fit sufficiently, reject H_0.

Use the same observations and outcome, an intercept in both models, and the same null hypothesis as in the coefficient formula.

Example: Are the coefficients on *strs* and *frpmfracs* zero?

Restricted population regression (that is, under H_0):

$$TestScore_i = \beta_0 + \beta_2 edfracbachplus_i + u_i$$

Unrestricted population regression (permits both H_0 and H_1):

$$TestScore_i = \beta_0 + \beta_1 frpmfracs_i + \beta_2 edfracbachplus_i + \beta_3 strs_i + u_i$$

The fit will be better (higher R^2) in the unrestricted regression. But by how much should it be higher for *strs* and *frpmfracs* to be deemed statistically significantly different from zero?

The Same F-statistic from R^2

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k_U - 1)} = \frac{(R_U^2 - R_R^2)/q}{(1 - R_U^2)/(n - k_U - 1)}.$$

- R and U denote restricted and unrestricted OLS fits.
- q is the number of restrictions; k_U counts unrestricted regressors, excluding the intercept.
- Use the same dataset and outcome, nested linear restrictions, and an intercept in both models.
- $SER^2 = SSR_U/(n - k_U - 1)$ is the variance estimate used in the coefficient formula.
- For $q = 2$, this is exactly the two-coefficient formula shown earlier.

A greater improvement in fit gives a larger F-statistic.

Example: Restricted and Unrestricted Fits

Restricted fitted regression:

$$\widehat{TestScore}_i = 705.157 + 203.582 \times edfracbachplus_i$$

with $R^2_R = 0.2320$.

Unrestricted fitted regression:

$$\begin{aligned}\widehat{TestScore}_i = 802.045 &- 128.685 \times frpmfracs_i \\ &+ 60.729 \times edfracbachplus_i + 0.564 \times strs_i\end{aligned}$$

with $R^2_U = 0.4775$.

Example: Comparing the Two School Regressions

$$F = \frac{(0.4775 - 0.2320)/2}{(1 - 0.4775)/(500 - 3 - 1)} = 116.5244.$$

- There are $q = 2$ restrictions and $500 - 3 - 1 = 496$ residual degrees of freedom.
- Under the maintained assumptions and conditionally normal errors, use $F_{2,496}$ exactly; otherwise use a large-sample approximation.
- The coefficient formula with the conventional covariance matrix gives the same F-statistic for these fits. The underlying data and covariance estimates are not provided here for a numerical verification.

Worked Example: One Dataset, Two Formulas

Constructed data: $n = 8$, $k_U = 2$. Test $H_0 : \beta_1 = \beta_2 = 0$ ($q = 2$).

i	X_1	X_2	Y
1	-1	-2	6
2	-1	-2	8
3	-1	0	8
4	-1	0	10
5	1	0	10
6	1	0	12
7	1	2	12
8	1	2	14

Unrestricted fit

$$\widehat{Y}_i = 10 + X_{1i} + X_{2i}.$$

Residuals: $(-1, 1, -1, 1, -1, 1, -1, 1)$; $SSR_U = 8$.

Restricted fit

$$\widehat{Y}_i = \bar{Y} = 10, \qquad SSR_R = TSS_Y = 48.$$

$$df = 8 - 2 - 1 = 5, \qquad SER^2 = \frac{8}{5}.$$

Same Example: Coefficient-Based F

The slope block of $\widehat{V} = SER^2(X'X)^{-1}$ is

$$\widehat{V}_s = \begin{pmatrix} 2/5 & -1/5 \\ -1/5 & 1/5 \end{pmatrix}.$$

$$t_1 = \frac{1}{\sqrt{2/5}} = \sqrt{5/2}, \qquad t_2 = \frac{1}{\sqrt{1/5}} = \sqrt{5}, \qquad \hat{\rho} = \frac{-1/5}{\sqrt{(2/5)(1/5)}} = -\frac{1}{\sqrt{2}}.$$

$$F = \frac{1}{2}\frac{(5/2) + 5 - 2(-1/\sqrt{2})\sqrt{5/2}\sqrt{5}}{1 - 1/2} = \boxed{12.5}.$$

The covariance term is essential: simply averaging t_1^2 and t_2^2 would give 3.75, not the joint F-statistic.

Same Example: Fit-Based F

$$R_U^2 = 1 - \frac{8}{48} = \frac{5}{6}, \qquad R_R^2 = 0.$$

$$F = \frac{(5/6 - 0)/2}{(1 - 5/6)/5} = 12.5, \qquad F = \frac{(48 - 8)/2}{8/5} = 12.5.$$

$$\boxed{F_{\text{coefficient formula}} = F_{\text{fit formula}} = 12.5}$$

With conditionally normal errors, the exact null reference is $F_{2,5}$:

$$p = \Pr(F_{2,5} \geq 12.5) = \left(\frac{1}{6}\right)^{5/2} = .01134023.$$

Both formulas give the same p-value and reject at 5%. Without normality, inference uses a large-sample approximation, which need not be accurate at $n = 8$.

Testing Single Restrictions on Multiple Coefficients

Consider the following specification

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

and the null and alternative hypothesis,

$$H_0 := \beta_1 = \beta_2 \quad \text{VS} \qquad H_1 \quad := \beta_1 \neq \beta_2$$

This null imposes a single restriction on multiple coefficients – it is not a joint hypothesis with multiple restrictions!

Here are two methods for testing single restrictions on multiple coefficients:

1. Rearrange the regressors so that the restriction becomes a restriction on a single coefficient in an equivalent regression; or,
2. Perform the test directly (Some software, including STATA, lets you test restrictions using multiple coefficients directly)

Rearranging

From the following specification

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

Add and substract $\beta_2 X_{1i}$:

$$\begin{aligned} Y_i &= \beta_0 + (\beta_1 - \beta_2) X_{1i} + \beta_2 (X_{2i} + X_{1i}) + u_i \\ &= \beta_0 + \gamma_1 X_{1i} + \beta_2 W_i + u_i \end{aligned}$$

Then test:

$$H_0 := \gamma_1 = 0 \quad \text{VS} \qquad H_1 \quad := \gamma_1 \neq 0$$

The original and rearranged regressions have the same R^2, the same predicted values, and the same residuals, but the problem is now a simpler one.

Testing Equality Using the Same Example

Test $H_0 : \beta_1 = \beta_2$, a single restriction ($q = 1$).

$$\hat{d} = \hat{\beta}_1 - \hat{\beta}_2 = 1 - 1 = 0.$$

$$\begin{aligned}\widehat{\mathrm{Var}}(\hat{d}) &= \widehat{\mathrm{Var}}(\hat{\beta}_1) + \widehat{\mathrm{Var}}(\hat{\beta}_2) - 2\widehat{\mathrm{Cov}}(\hat{\beta}_1, \hat{\beta}_2) \\ &= .4 + .2 - 2(-.2) = 1.\end{aligned}$$

$$SE(\hat{d}) = 1, \qquad t = \frac{0}{1} = 0, \qquad F = t^2 = 0.$$

Under the exact normal-error reference (t_5 or $F_{1,5}$), $p = 1$: do not reject equality. The same conventional covariance matrix is used throughout.

Confidence Sets for Multiple Coefficients

Consider the following regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

What is a joint confidence set for β_1 and β_2?

A 95% joint confidence set is:

- A set-valued function of the data that contains the true coefficient(s) in 95% hypothetical repeated samples.
- Equivalently, the set of coefficient values that cannot be rejected at the 5% significance level

You can find a 95% confidence set as the set of (β_1, β_2) that cannot be rejected at the 5% level using an F-test (why not just combine the two 95% confidence intervals?).

Confidence Sets for Multiple Coefficients

- Let $F(\beta_{1,0}, \beta_{2,0})$ test $H_0 : \beta_1 = \beta_{1,0},\ \beta_2 = \beta_{2,0}$ using the conventional covariance matrix.
- The 95% confidence set is

$$\{(\beta_{1,0}, \beta_{2,0}) : F(\beta_{1,0}, \beta_{2,0}) < c\}, \qquad c = F_{2,n-k-1;.95}.$$

- For large denominator degrees of freedom, $c \approx 3.00$.
- Under conditionally normal errors, coverage is exactly 95%; otherwise it is a large-sample approximation.
- Inverting a 5% test includes precisely the coefficient pairs that are not rejected.

Confidence Sets for Multiple Coefficients

For each candidate pair, define

$$t_1 = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)}, \qquad t_2 = \frac{\hat{\beta}_2 - \beta_{2,0}}{SE(\hat{\beta}_2)}.$$

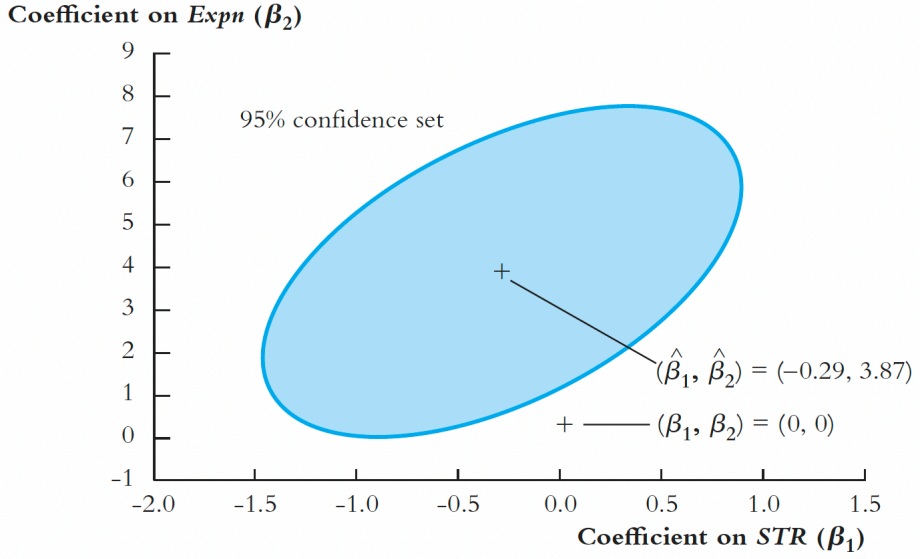
The confidence set uses the same coefficient formula:

$$\left\{ (\beta_{1,0}, \beta_{2,0}) : \frac{t_1^2 + t_2^2 - 2\hat{\rho} t_1 t_2}{2(1-\hat{\rho}^2)} < c \right\}, \qquad c = F_{2,n-k-1;.95}.$$

The cross term is $-2\hat{\rho} t_1 t_2$. The boundary $F = c$ is an ellipse.

For large denominator degrees of freedom, $c \approx 3.00$. The next figure illustrates the joint region and the separate intervals.

Confidence Sets for Multiple Coefficients



How to decide what variables to include

1. Identify the variable of interest
2. Think of the omitted causal effects that could result in omitted variable bias
3. Include those omitted causal effects if you can or, if you can't, include variables correlated with them that serve as control variables.
   - The control variables are effective if the conditional mean independence assumption plausibly holds, that is, if u is uncorrelated with the share of subsidized meals once the control variables are included.
   - This results in a "base" or "benchmark" model.
4. Also specify a range of plausible alternative models, which include additional candidate variables.
5. Estimate your base model and plausible alternative specifications ("sensitivity checks").
   - Does a candidate variable change the coefficient of interest β_1?
   - Is a candidate variable statistically significant?
   - Use judgment, not a mechanical recipe...
   - Don't just try to maximize the R^2 !

What the R^2 tells you, and what it doesn't

It is easy to fall into the trap of maximizing the R^2 and $\overline{R}^2$ but this loses sight of our real objective: an unbiased estimator of the class size effect.

- A high R^2 (or $\overline{R}^2$) means that the regressors explain the variation in your data
- A high R^2 (or $\overline{R}^2$) **does not** mean that you have eliminated omitted variable bias
- A high R^2 (or $\overline{R}^2$) **does not** mean that you have an unbiased estimator of a causal effect β_1
- A high R^2 (or $\overline{R}^2$) **does not** mean that the included variables are statistically significant — this must be determined using hypotheses tests.

Summary

- Multiple regression allows you to estimate the effect on Y of a change in X_1, holding other included variables constant.
- If you can measure a variable, you can avoid omitted variable bias from that variable by including it.
- If you can't measure the omitted variable, you still might be able to control for its effect by including a control variable.
- There is no simple recipe for deciding which variables belong in a regression – you must exercise judgment.
- One approach is to specify a base model – relying on a priori reasoning – then explore the sensitivity of the key estimate(s) in alternative specifications.

Material I

- **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 7